GridLayout

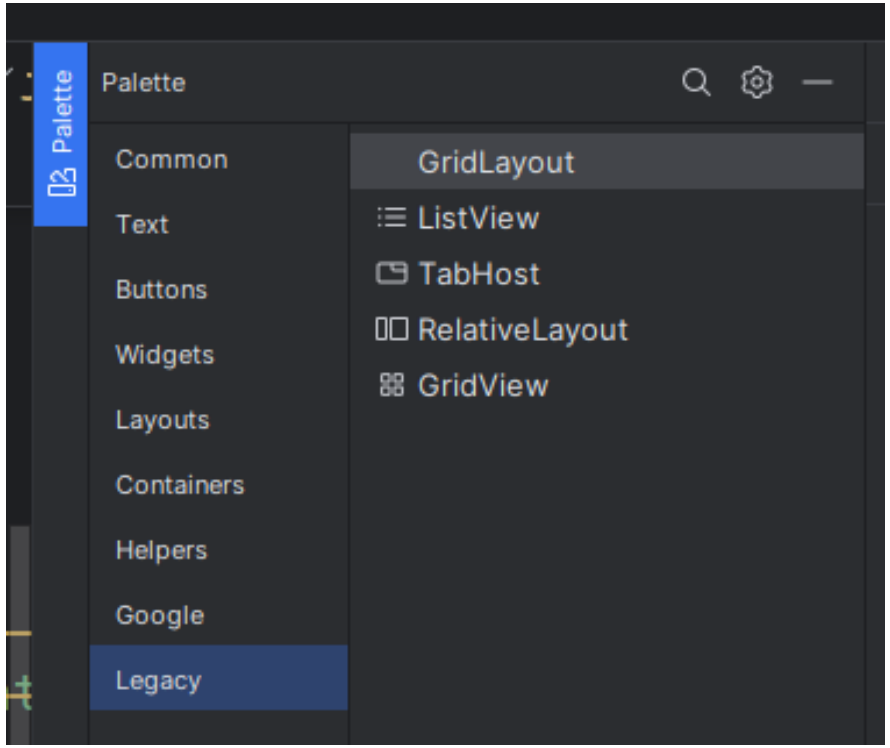
Android Studio

GridLayout

raw (3 X 3) column

column

	0	1	2
raw 0	0 , 0	0 , 1	0 , 2
raw 1	1 , 0	1 , 1	1 , 2
raw 2	2, 0	2 , 1	2 , 2



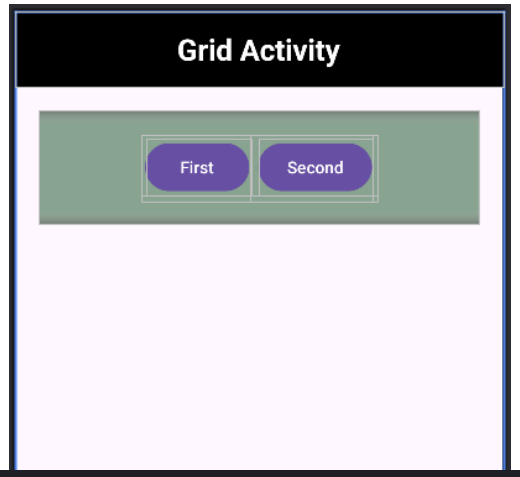
GridLayout Via GridView

- **GridLayout** is a ViewGroup (layout manager) for arranging a *fixed*, predetermined number of child views in a grid structure, often defined directly in XML. It's suitable for simple, static grids like a calculator interface.

- **GridView** is an AdapterView used to display a dynamic collection of items (from a database, array, or list) in a two-dimensional, scrollable grid. It requires an **Adapter** to feed data and views efficiently, reusing and recycling views for performance (like a gallery of images).

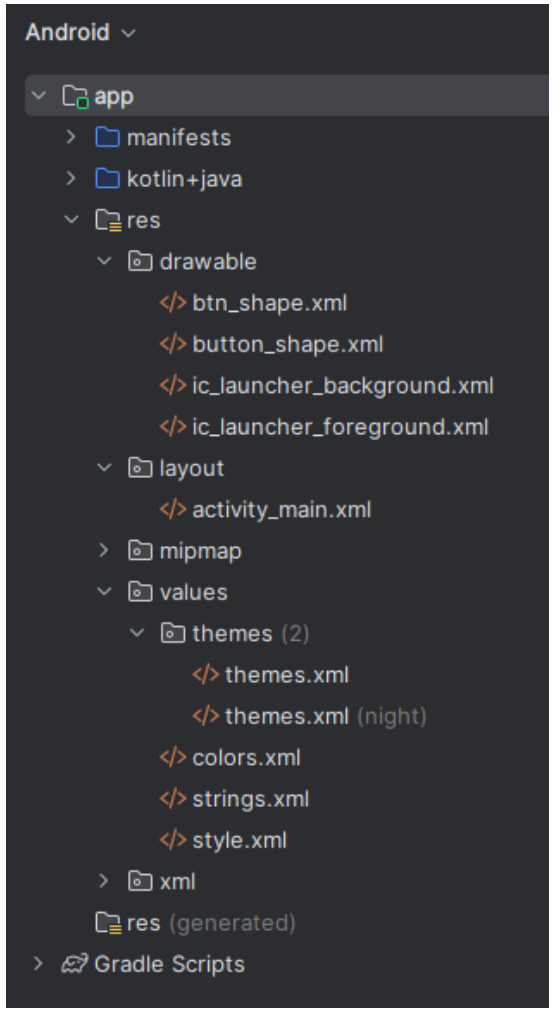
Project 1

```
<androidx.gridlayout.widget.GridLayout
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    app:rowCount="1"
    app:columnCount="2"
    app:useDefaultMargins="true">
    <Button
        android:id="@+id/first"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="First"
        app:layout_column="0"
        app:layout_row="0" />
    <Button
        android:id="@+id/second"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Second"
        app:layout_column="1"
        app:layout_row="0" />
</androidx.gridlayout.widget.GridLayout>
```



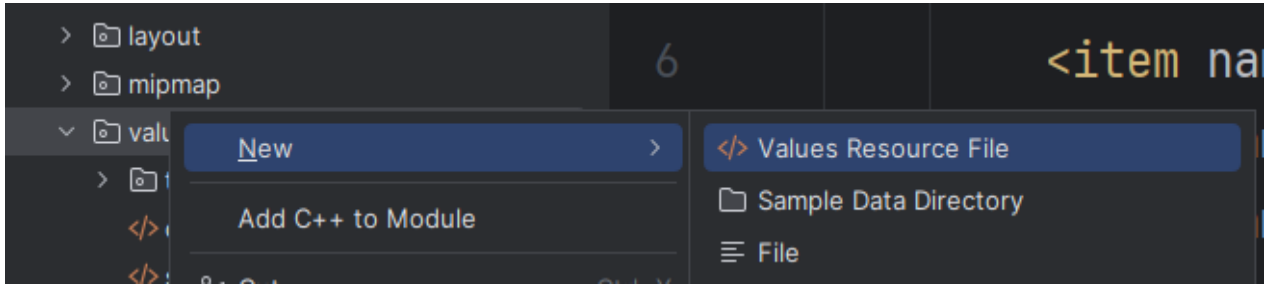

```
val first = findViewById<Button>( id = R.id.first)
val second = findViewById<Button>( id = R.id.second)

first.setOnClickListener {
    val i = Intent( packageContext = this,  cls = FirstActivity::class.java)
    startActivity( intent = i)
}

second.setOnClickListener {
    val i = Intent( packageContext = this,  cls = SecondActivity::class.java)
    startActivity( intent = i)
}
```

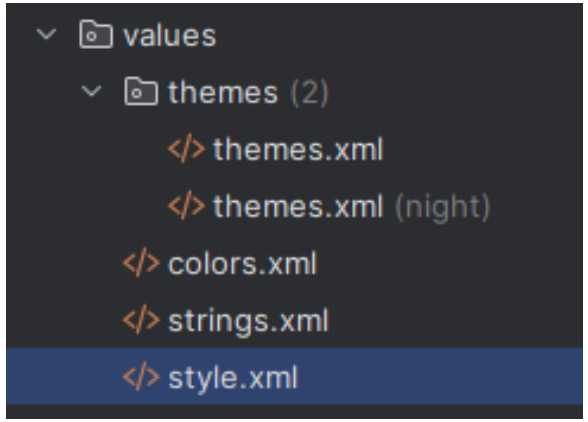
Project 2

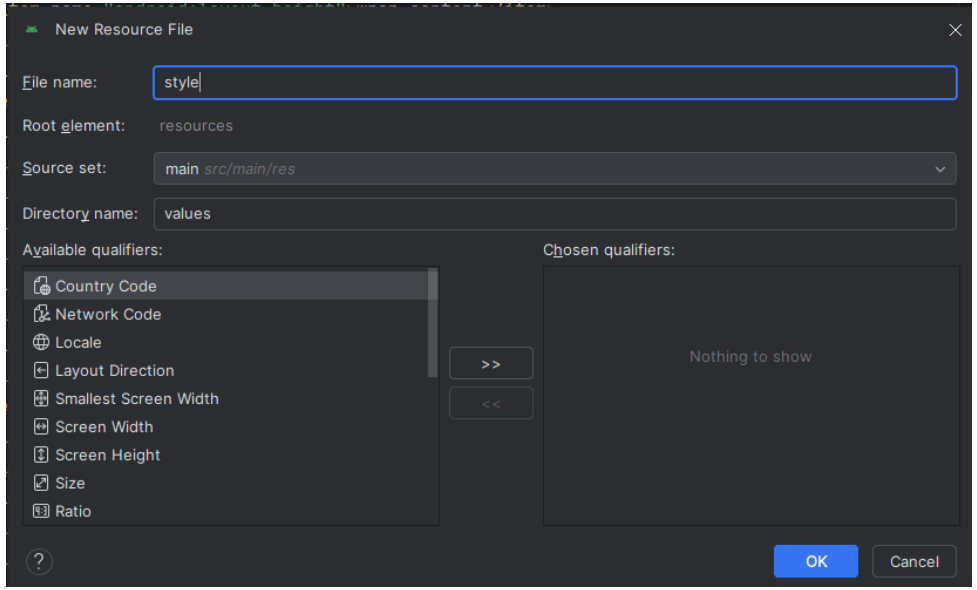

Android
app
manifests
kotlin+java
res
drawable
btn_shape.xml
button_shape.xml
ic_launcher_background.xml
ic_launcher_foreground.xml
layout
activity_main.xml
mipmap
values
themes (2)
themes.xml
themes.xml (night)
colors.xml
strings.xml
style.xml
xml
res (generated)
Gradle Scripts

Grid Activity
BUTTON
BUTTON
BUTTON
BUTTON
BUTTON
BUTTON
BUTTON
BUTTON
BUTTON
First Name
Enter the name
Last Name
Enter the name
START

Create new values (Style.xml)





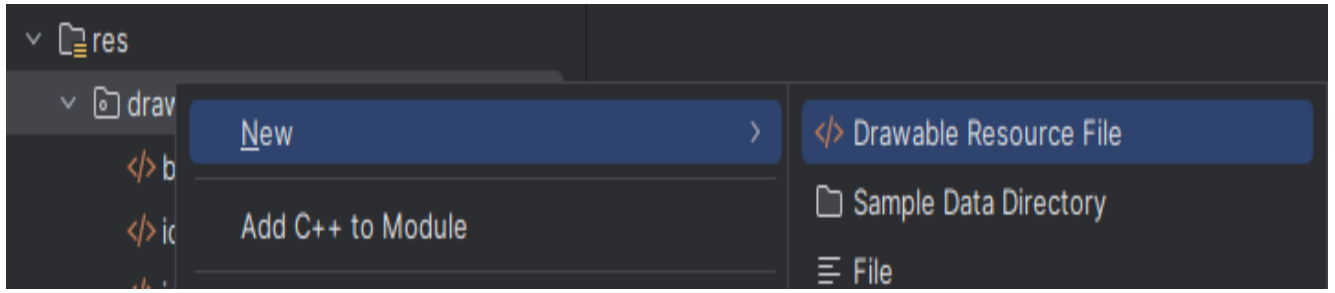


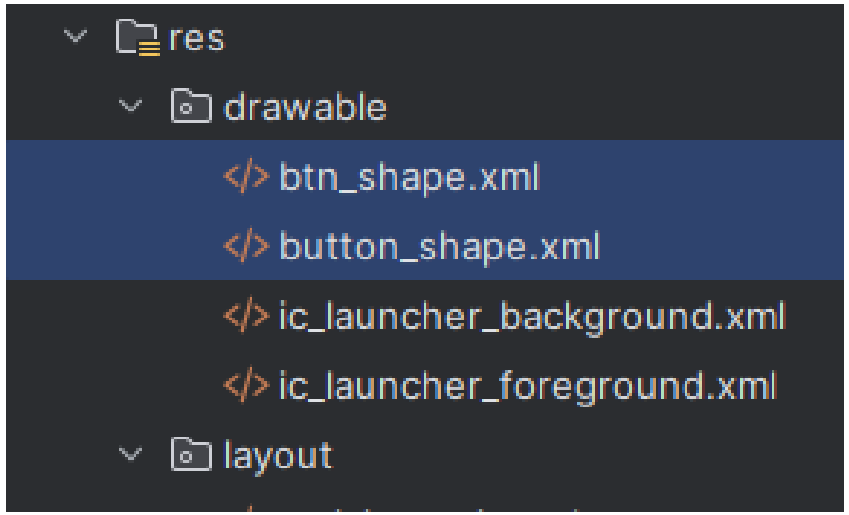
Style.xml

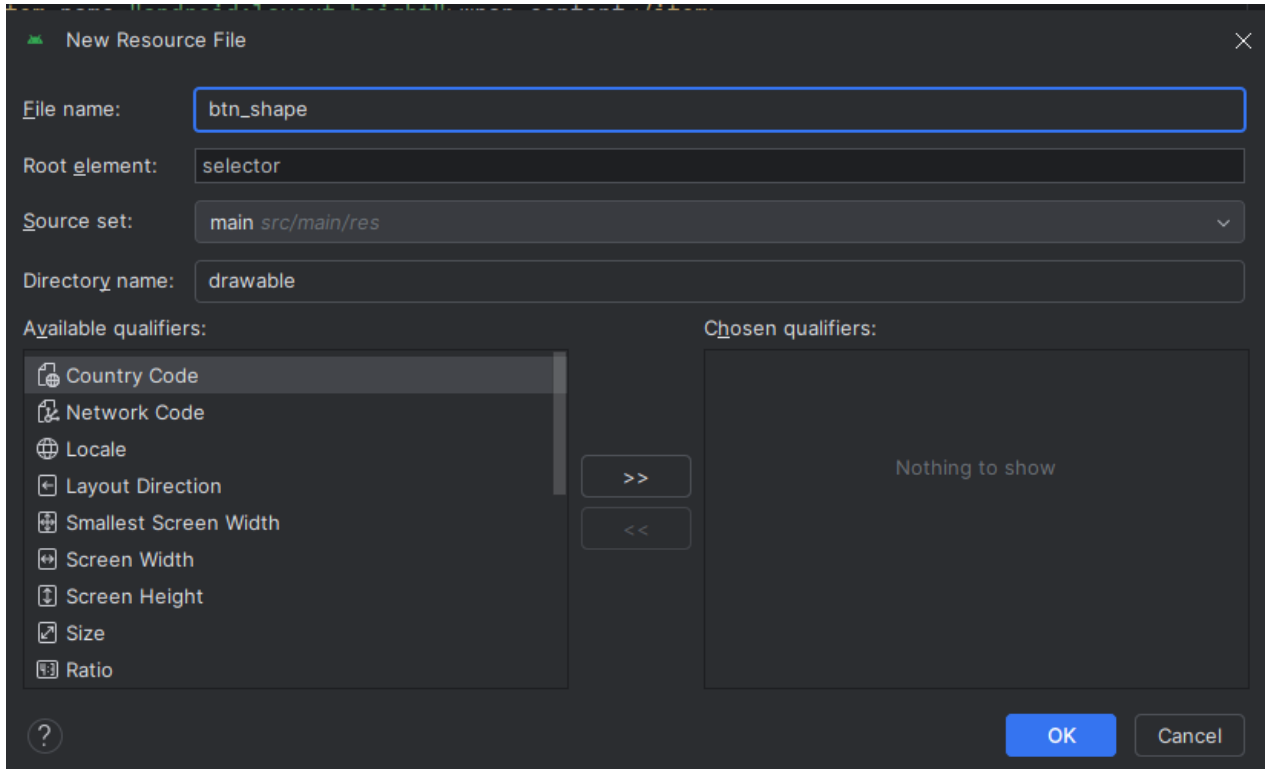
style.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <style name="btnstyle" >
        <item name="android:padding">10dp</item>
        <item name="backgroundTint">#2A95C7</item>
        <item name="android:textSize">15dp</item>
        <item name="android:layout_width">90dp</item>
        <item name="android:layout_height">wrap_content</item>
        <item name="android:layout_margin">10dp</item>
    </style>
    <style name="tvstyle" >
        <item name="android:padding">7dp</item>
        <item name="background">#2A95C7</item>
        <item name="android:textSize">20dp</item>
        <item name="android:textColor">@color/black</item>
        <item name="android:textStyle">bold</item>
        <item name="android:layout_width">wrap_content</item>
        <item name="android:layout_height">wrap_content</item>
        <item name="android:layout_margin">10dp</item>
    </style>
    <style name="ptstyle" >
        <item name="android:padding">7dp</item>
        <item name="background">#9FB8C5</item>
        <item name="android:inputType">text</item>
        <item name="android:textSize">20dp</item>
        <item name="android:hint">Enter the name</item>
        <item name="android:textColor">@color/black</item>
        <item name="android:textStyle">bold</item>
        <item name="android:layout_width">wrap_content</item>
        <item name="android:layout_height">wrap_content</item>
        <item name="android:layout_margin">10dp</item>
        <item name="android:ems">10</item>
    </style>
</resources>
```

Button Shape (background) add two files







Update new drawable files

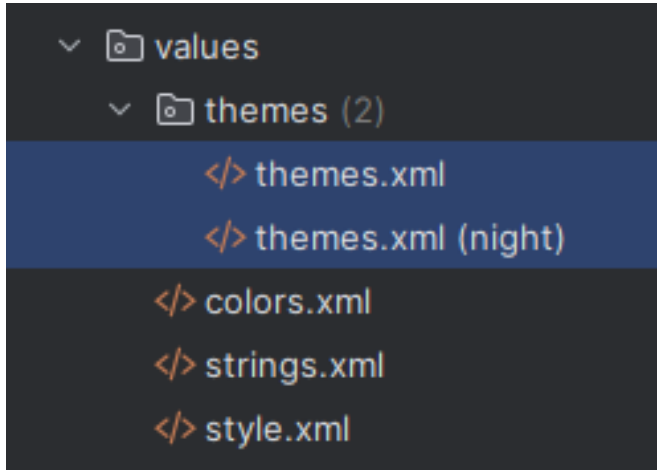
button_shape.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
        <solid android:color="#277FA6" />
        <corners android:radius="100dp" />
        <stroke android:color="@color/black" android:width="3dp" />
</shape>
```

button_shape.xml btn_shape.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<selector xmlns:android="http://schemas.android.com/apk/res/android">
<item>
    <shape android:shape="rectangle" >
        <solid android:color="#068E80" />
        <stroke android:color="@color/black" android:width="3dp" />
        <corners android:radius="100dp" />
    </shape>
</item>
</selector>
```

Update two Themes.xml file



values\themes.xml × night\themes.xml

```xml
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.GridApp1" parent="Theme.AppCompat.NoActionBar">
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>

    <style name="Theme.GridApp1" parent="Base.Theme.GridApp1" />
</resources>
```

style.xml night\themes.xml ×

```xml
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.GridApp1" parent="Theme.AppCompat.NoActionBar">
        <!-- Customize your dark theme here. -->
        <!-- <item name="colorPrimary">@color/my_dark_primary</item> -->
    </style>
</resources>
```

activity_main.xml

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#CCCCCC"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@color/black"
        android:gravity="center"
        android:padding="15dp"
        android:text="Grid Activity"
        android:textColor="@color/white"
        android:textSize="30sp"
        android:textStyle="bold"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
```

```xml
    <androidx.gridlayout.widget.GridLayout
        android:id="@+id/gridLayout"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="16dp"
        android:background="#A29795"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView">

        <Button
            android:id="@+id/button"
            android:background="@drawable/button_shape"
            android:layout_margin="10dp"
            android:text="Button"
            app:layout_column="0"
            app:layout_row="0" />

        <Button
            android:id="@+id/button2"
            android:background="@drawable/button_shape"
            android:layout_margin="10dp"

            android:text="Button"
            app:layout_column="1"
            app:layout_row="0" />

        <Button
            android:id="@+id/button3"
            android:layout_margin="10dp"
            android:background="@drawable/button_shape"
            android:text="Button"
            app:layout_column="2"
            app:layout_row="0" />
```

```xml
        <Button
            android:id="@+id/button4"
            android:text="Button"
            android:background="@drawable/btn_shape"
            android:layout_margin="10dp"
            app:layout_column="0"
            app:layout_row="1" />
        <Button
            android:id="@+id/button5"
            android:background="@drawable/btn_shape"
            android:layout_margin="10dp"
            android:text="Button"
            app:layout_column="1"
            app:layout_row="1" />
        <Button
            android:id="@+id/button6"
            android:background="@drawable/btn_shape"
            android:layout_margin="10dp"
            android:text="Button"
            app:layout_column="2"
            app:layout_row="1" />
        <Button
            android:id="@+id/button7"
            android:text="Button"
            app:layout_column="0"
            app:layout_row="2" />
        <Button
            android:id="@+id/button8"
            android:text="Button"
            app:layout_column="1"
            app:layout_row="2" />
        <Button
            android:id="@+id/button9"
            android:text="Button"
            app:layout_column="2"
            app:layout_row="2" />
    </androidx.gridlayout.widget.GridLayout>
```

activity_main.xml

```
<androidx.gridlayout.widget.GridLayout
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="40dp"
    android:background="#897E7E"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/gridLayout">
    <TextView
        android:id="@+id/textView2"
        style="@style/tvstyle"
        android:text="First Name"
        app:layout_row="0"
        app:layout_column="0"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
    <EditText
        android:id="@+id/editTextText"
        style="@style/ptstyle"
        app:layout_column="1"
        app:layout_row="1" />
```

```
    <TextView
        android:id="@+id/textView3"
        style="@style/tvstyle"
        android:text="Last Name"
        app:layout_column="0"
        app:layout_row="2" />
    <EditText
        android:id="@+id/editTextText2"
        style="@style/ptstyle"
        app:layout_column="1"
        app:layout_row="3" />
    <Button
        android:id="@+id/start"
        android:background="@drawable/button_shape"
        android:text="Start"
        android:backgroundTint="@color/black"
        app:layout_column="1"
        app:layout_row="4" />
</androidx.gridlayout.widget.GridLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```